

Gibson–Schwartz PINN: Pricing Derivation and Loss Architecture

Evan Lynch

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1 Setup under the physical measure $\mathbb{P}$

Work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, with spot price S_t and convenience yield δ_t evolving as

$$dS_t = (\mu - \delta_t)S_t dt + \sigma_1 S_t dW_1^{\mathbb{P}}, \quad (1)$$

$$d\delta_t = \kappa(\alpha^{\mathbb{P}} - \delta_t) dt + \sigma_2 dW_2^{\mathbb{P}}, \quad (2)$$

with correlated drivers, $dW_1^{\mathbb{P}} dW_2^{\mathbb{P}} = \rho dt$, $\rho \in (-1, 1)$.

2 Change to the risk-neutral measure $\mathbb{Q}$

Under $\mathbb{Q}$ the traded asset must drift at the risk-free rate r , so we introduce a market price of risk λ_i for each driver and seek a measure under which

$$dW_i^{\mathbb{Q}} = dW_i^{\mathbb{P}} + \lambda_i dt, \quad i = 1, 2. \quad (3)$$

Since $W_1^{\mathbb{P}}$ and $W_2^{\mathbb{P}}$ are correlated, the change of measure is constructed on an orthogonal basis. Let B_1, B_2 be independent $\mathbb{P}$ -Brownian motions with

$$W_1^{\mathbb{P}} = B_1, \quad W_2^{\mathbb{P}} = \rho B_1 + \sqrt{1 - \rho^2} B_2. \quad (4)$$

Requiring the shifts in Equation (3) fixes the Girsanov kernel (θ_1, θ_2) through $\theta_1 = \lambda_1$ and $\rho\theta_1 + \sqrt{1 - \rho^2}\theta_2 = \lambda_2$, so that

$$\theta_1 = \lambda_1, \quad \theta_2 = \frac{\lambda_2 - \rho\lambda_1}{\sqrt{1 - \rho^2}}, \quad (5)$$

and by Girsanov's theorem the measure change is defined by the Radon–Nikodým derivative

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left(- \int_0^t \theta_1 dB_1 - \int_0^t \theta_2 dB_2 - \frac{1}{2} \int_0^t (\theta_1^2 + \theta_2^2) ds \right), \quad (6)$$

with

$$\theta_1^2 + \theta_2^2 = \frac{\lambda_1^2 - 2\rho\lambda_1\lambda_2 + \lambda_2^2}{1 - \rho^2}. \quad (7)$$

For $\rho = 0$ this reduces to the familiar $\exp \left(- \int \lambda_1 dW_1^{\mathbb{P}} - \int \lambda_2 dW_2^{\mathbb{P}} - \frac{1}{2} \int (\lambda_1^2 + \lambda_2^2) ds \right)$; for $\rho \neq 0$ the cross term in Equation (7) is essential, since the quadratic variation of the exponent picks up the covariation of the drivers. The dynamics below are stated directly in terms of $dW_i^{\mathbb{P}} = dW_i^{\mathbb{Q}} - \lambda_i dt$ and are unaffected by the choice of basis.

2.1 Substituting the measure change

Rearranging Equation (3) gives $dW_i^{\mathbb{P}} = dW_i^{\mathbb{Q}} - \lambda_i dt$. Substituting into Equations (1) and (2):

$$dS_t = [(\mu - \delta_t) - \sigma_1 \lambda_1] S_t dt + \sigma_1 S_t dW_1^{\mathbb{Q}}, \quad (8)$$

$$d\delta_t = (\kappa \alpha^{\mathbb{P}} - \sigma_2 \lambda_2 - \kappa \delta_t) dt + \sigma_2 dW_2^{\mathbb{Q}}. \quad (9)$$

These are correct but leave λ_1, λ_2 unspecified, and the convenience-yield drift is no longer in mean-reverting form. We resolve each in turn.

2.2 Pinning down λ_1 : the no-arbitrage drift condition

The economic content of no-arbitrage here is the cost-of-carry condition. In Equation (1), μ is the total expected return on holding the physical commodity: the price drift is $\mu - \delta_t$, and the holder additionally collects the convenience yield δ_t as a non-pecuniary dividend (avoiding stock-outs, retaining operational flexibility), the two summing to μ . Under $\mathbb{Q}$, no-arbitrage requires the total expected return on the traded spot to equal the risk-free rate r . Since the convenience yield accrues to the holder like a dividend, this is equivalent to a risk-neutral price drift of $r - \delta_t$. Matching this to the drift in Equation (8),

$$(\mu - \delta_t) - \sigma_1 \lambda_1 = r - \delta_t \implies \lambda_1 = \frac{\mu - r}{\sigma_1}, \quad (10)$$

which identifies λ_1 as the market price of spot risk (the Sharpe ratio of the spot process). Substituting back:

$$dS_t = (r - \delta_t) S_t dt + \sigma_1 S_t dW_1^{\mathbb{Q}}. \quad (11)$$

2.3 Pinning down $\alpha^{\mathbb{Q}}$: restoring mean-reverting form

Factoring κ back out of the drift in Equation (9):

$$\kappa \alpha^{\mathbb{P}} - \sigma_2 \lambda_2 = \kappa \left(\alpha^{\mathbb{P}} - \frac{\sigma_2 \lambda_2}{\kappa} \right) =: \kappa \alpha^{\mathbb{Q}}. \quad (12)$$

Unlike λ_1 , the convenience-yield risk premium λ_2 is not fixed by a no-arbitrage argument, since δ_t is not a traded asset; it is identified jointly with the other parameters by calibration to the observed futures term structure. Substituting back:

$$d\delta_t = \kappa (\alpha^{\mathbb{Q}} - \delta_t) dt + \sigma_2 dW_2^{\mathbb{Q}}, \quad \alpha^{\mathbb{Q}} = \alpha^{\mathbb{P}} - \frac{\sigma_2 \lambda_2}{\kappa}. \quad (13)$$

Only the long-run level shifts ($\alpha^{\mathbb{P}} \rightarrow \alpha^{\mathbb{Q}}$); κ and σ_2 are unchanged.

3 Itô's lemma and the futures PDE

Let $F = F(S_t, \delta_t, t)$ be the futures price. By the bivariate Itô lemma,

$$dF = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial S} dS_t + \frac{\partial F}{\partial \delta} d\delta_t + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (dS_t)^2 + \frac{\partial^2 F}{\partial S \partial \delta} (dS_t)(d\delta_t) + \frac{1}{2} \frac{\partial^2 F}{\partial \delta^2} (d\delta_t)^2. \quad (14)$$

The quadratic-variation terms, with $dW_1^{\mathbb{Q}} dW_2^{\mathbb{Q}} = \rho dt$, are

$$(dS_t)^2 = \sigma_1^2 S_t^2 dt, \quad (d\delta_t)^2 = \sigma_2^2 dt, \quad (dS_t)(d\delta_t) = \rho \sigma_1 \sigma_2 S_t dt. \quad (15)$$

Substituting the $\mathbb{Q}$ -dynamics gives

$$dF = \left[\frac{\partial F}{\partial t} + (r - \delta_t) S_t \frac{\partial F}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta_t) \frac{\partial F}{\partial \delta} + \frac{1}{2} \sigma_1^2 S_t^2 \frac{\partial^2 F}{\partial S^2} + \rho \sigma_1 \sigma_2 S_t \frac{\partial^2 F}{\partial S \partial \delta} + \frac{1}{2} \sigma_2^2 \frac{\partial^2 F}{\partial \delta^2} \right] dt + \sigma_1 S_t \frac{\partial F}{\partial S} dW_1^{\mathbb{Q}} + \sigma_2 \frac{\partial F}{\partial \delta} dW_2^{\mathbb{Q}}. \quad (16)$$

A futures contract requires no initial investment, so F must be a $\mathbb{Q}$ -martingale: its drift vanishes, $\mathbb{E}^{\mathbb{Q}}[dF] = 0$. Setting the bracket to zero gives the Gibson–Schwartz PDE,

$$\frac{\partial F}{\partial t} + (r - \delta) S \frac{\partial F}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta) \frac{\partial F}{\partial \delta} + \frac{1}{2} \sigma_1^2 S^2 \frac{\partial^2 F}{\partial S^2} + \rho \sigma_1 \sigma_2 S \frac{\partial^2 F}{\partial S \partial \delta} + \frac{1}{2} \sigma_2^2 \frac{\partial^2 F}{\partial \delta^2} = 0, \quad (17)$$

subject to the terminal condition $F(S_T, \delta_T, T) = S_T$.

4 Change of variables: time-to-maturity τ

In calendar time the terminal condition sits at $t = T$, which differs across contracts, undesirable for a network learning a single function valid across all maturities. Re-expressing in time-to-maturity $\tau = T - t$ collapses the terminal condition to the fixed point $\tau = 0$.

Let $\hat{F}(S, \delta, \tau) := F(S, \delta, t)$. Since $t = T - \tau$, the chain rule gives $\partial t / \partial \tau = -1$, so

$$\frac{\partial F}{\partial t} = -\frac{\partial \hat{F}}{\partial \tau}. \quad (18)$$

The spatial derivatives are unchanged, since S and δ depend on neither t nor τ . Substituting Equation (18) into Equation (17) and rearranging yields a forward equation in τ ,

$$\frac{\partial \hat{F}}{\partial \tau} = (r - \delta) S \frac{\partial \hat{F}}{\partial S} + \kappa(\alpha^{\mathbb{Q}} - \delta) \frac{\partial \hat{F}}{\partial \delta} + \frac{1}{2} \sigma_1^2 S^2 \frac{\partial^2 \hat{F}}{\partial S^2} + \rho \sigma_1 \sigma_2 S \frac{\partial^2 \hat{F}}{\partial S \partial \delta} + \frac{1}{2} \sigma_2^2 \frac{\partial^2 \hat{F}}{\partial \delta^2}, \quad (19)$$

with the terminal condition becoming an initial condition,

$$\hat{F}(S, \delta, 0) = S. \quad (20)$$

This is a well-posed initial value problem and is the form used in the PINN: collocation points are sampled over (S, δ, τ) , with τ the interval between each observation date and the contract's delivery date.

5 Closed-form futures price

Equation (19) with $\hat{F}(S, \delta, 0) = S$ admits a closed-form, exponential-affine solution. Take the ansatz

$$\hat{F}(\tau, S, \delta) = S \exp(B(\tau) \delta + A(\tau)), \quad (21)$$

whose derivatives are

$$\frac{\partial \hat{F}}{\partial S} = \frac{\hat{F}}{S}, \quad \frac{\partial^2 \hat{F}}{\partial S^2} = 0, \quad \frac{\partial \hat{F}}{\partial \delta} = B \hat{F}, \quad \frac{\partial^2 \hat{F}}{\partial \delta^2} = B^2 \hat{F}, \quad \frac{\partial^2 \hat{F}}{\partial S \partial \delta} = B \frac{\hat{F}}{S}, \quad \frac{\partial \hat{F}}{\partial \tau} = \hat{F}(B' \delta + A'). \quad (22)$$

Substitution. Inserting these into Equation (19) and dividing by $\hat{F}$:

$$B'\delta + A' = (r - \delta) + \kappa(\alpha^{\mathbb{Q}} - \delta)B + \rho\sigma_1\sigma_2B + \frac{1}{2}\sigma_2^2B^2. \quad (23)$$

Matching the coefficient of δ and the constant term (the $\frac{1}{2}\sigma_1^2S^2$ term drops out, since $\hat{F}_{SS} = 0$) gives two decoupled ODEs with $B(0) = A(0) = 0$:

$$B'(\tau) = -1 - \kappa B(\tau), \quad (24)$$

$$A'(\tau) = r + (\kappa\alpha^{\mathbb{Q}} + \rho\sigma_1\sigma_2)B(\tau) + \frac{1}{2}\sigma_2^2B(\tau)^2. \quad (25)$$

Solving for $B(\tau)$. The linear ODE $B' + \kappa B = -1$ has integrating factor $e^{\kappa\tau}$, giving $e^{\kappa\tau}B = -\frac{1}{\kappa}e^{\kappa\tau} - C$. Applying $B(0) = 0$ fixes $C = -\frac{1}{\kappa}$, so

$$B(\tau) = -\frac{1 - e^{-\kappa\tau}}{\kappa}. \quad (26)$$

Solving for $A(\tau)$. Squaring,

$$B(\tau)^2 = \frac{1 - 2e^{-\kappa\tau} + e^{-2\kappa\tau}}{\kappa^2}; \quad (27)$$

the $e^{-2\kappa\tau}$ term is the source of the second exponential in the result below. Substituting B and B^2 into A' and integrating term by term from 0 to τ (using $A(0) = 0$):

$$A(\tau) = \left(r - \alpha^{\mathbb{Q}} + \frac{1}{2}\frac{\sigma_2^2}{\kappa^2} - \frac{\sigma_1\sigma_2\rho}{\kappa}\right)\tau + \frac{1}{4}\sigma_2^2\frac{1 - e^{-2\kappa\tau}}{\kappa^3} + \left(\alpha^{\mathbb{Q}}\kappa + \sigma_1\sigma_2\rho - \frac{\sigma_2^2}{\kappa}\right)\frac{1 - e^{-\kappa\tau}}{\kappa^2}. \quad (28)$$

Result.

$$\hat{F}(\tau, S, \delta) = S \exp(B(\tau)\delta + A(\tau)), \quad B(\tau) = -\frac{1 - e^{-\kappa\tau}}{\kappa}, \quad (29)$$

with $A(\tau)$ as in Equation (28). This matches the Schwartz (1997) Model 2 closed form term by term, under the identification $\hat{\alpha} = \alpha^{\mathbb{Q}}$ (his raw λ premium absorbed into $\alpha^{\mathbb{Q}} = \alpha^{\mathbb{P}} - \sigma_2\lambda_2/\kappa$, his T playing the role of τ).¹

Role of the closed form. The exponential-affine solution is not used in training. It serves two purposes: as the data-generating process for the Monte Carlo simulation study, and as an exact out-of-sample benchmark against which the trained surface and the recovered convenience-yield path are evaluated, a validation channel unavailable in settings whose pricing PDE admits no closed form. The reason it must be excluded from the objective is structural and is set out in Section 6.

6 Network parameterisation

The learning problem carries two unknowns of different type: the futures pricing function, a deterministic map on the state space, and the convenience-yield path, a latent state realised along the observed history. We parameterise them separately:

$$\hat{F}_\theta : (S, \delta, \tau) \mapsto \mathbb{R}_+, \quad \hat{\delta}_\phi : t \mapsto \mathbb{R}, \quad (30)$$

¹Schwartz (1997), eqs. (18)–(20), attributed to Jamshidian & Fein (1990) and Bjerk Sund (1991).

with parameter sets θ and ϕ trained jointly. The map $\hat{F}_\theta$ plays the role of the surface Φ_θ in Hoshisashi et al. (2024): it is the object on which the PDE residual, the initial condition, and the differential structure of the problem act, evaluated on mesh grids over (S, δ, τ) . The path $\hat{\delta}_\phi$ has no analogue in the implied-volatility setting and is the state-estimation content of this work: it enters the objective only through the data-misfit loss, where it selects the point on the surface at which each observed futures price is priced.

Two remarks fix conventions. First, $\hat{\delta}_\phi$ takes t alone as input. Admitting S_t as an input would let the network launder price information into the yield estimate and blur its interpretation as a filtered state; the correlation between the two factors is already carried by ρ in the dynamics that $\hat{\mathcal{L}}_f$ enforces. Second, the closed-form solution of Section 5 is excluded from training. Because it solves the pricing PDE identically for every value of the coordinate δ , composing it with $\hat{\delta}_\phi$ would render $\hat{\mathcal{L}}_f$ and $\hat{\mathcal{L}}_b$ identically zero and reduce the method to penalised least squares. This is a structural difference from the implied-volatility case, where the Black–Scholes map composed with a free volatility surface does not solve the Dupire equation and the residual retains content. Here the PDE constrains only a surface that is free to violate it; the closed form is reserved as the benchmark against which that surface is judged.

7 Inequality constraints: the $\hat{\mathcal{L}}_h$ terms

The data-misfit loss $\hat{\mathcal{L}}_t$ and the PDE-residual loss $\hat{\mathcal{L}}_f$ together pin the network output to the observed term structure and to the Gibson–Schwartz dynamics. Neither, however, constrains the inferred convenience yield $\hat{\delta}_\phi$ to lie in an economically admissible range: a path $\hat{\delta}_\phi$ may fit the observed futures and satisfy the pricing PDE while still taking values no rational holder of the physical commodity would tolerate. The inverse problem is in this sense ill-posed, and the inequality losses $\hat{\mathcal{L}}_h$ act as a regularisation device that restores admissibility.

We group the constraints into two families. Model-free constraints follow from static no-arbitrage alone and hold independently of the Gibson–Schwartz specification; they bound the futures price, or the convenience yield, pointwise at a given maturity τ . The economic-plausibility constraint instead bounds the latent yield directly. Each constraint $g(\cdot) \geq 0$ is enforced through a one-sided hinge penalty,

$$\hat{\mathcal{L}}_h = \frac{1}{N} \sum_{i=1}^N [\max(0, -g(x_i))]^2, \quad (31)$$

zero where the constraint holds and growing quadratically with the magnitude of any violation.

Domain of enforcement. The inequality penalties differ from those of the implied-volatility setting in where they may legitimately be imposed. There, the no-arbitrage conditions constrain the sign of derivatives of the surface in its own coordinates and are therefore enforced on dense mesh grids. Here the carry bounds constrain the observed market price given (S, τ) alone: the cash-and-carry ceiling of Section 7.1 makes no reference to the coordinate δ , and by Equation (35) the true Gibson–Schwartz surface violates it wherever δ lies below the threshold $((r + u)\tau - A(\tau))/B(\tau)$. Enforcing the ceiling over the full mesh would therefore contradict the PDE residual on part of the domain. The carry penalties are accordingly evaluated on the data manifold, at the inferred states

$$x_i = (S_i, \hat{\delta}_\phi(t_i), \tau_i), \quad i = 1, \dots, N_t, \quad (32)$$

while the convenience-yield floor of Section 7.3 constrains the latent path directly.

7.1 Model-free constraint I: cash-and-carry upper bound

The cash-and-carry strategy enforces an upper bound on the futures price. An arbitrageur who observes a futures price exceeding the cost of carrying the physical commodity to delivery can, at time t , borrow S_t at the risk-free rate, buy one unit of the commodity, pay the storage and insurance cost of holding it, and simultaneously sell the futures. At delivery the commodity is handed over against the futures price, the loan is repaid, and the strategy locks in a riskless profit if

$$F_t > S_t e^{(r+u)\tau}, \quad (33)$$

where $u \geq 0$ is the per-unit, per-unit-time cost of physical storage (including insurance), expressed as a continuously compounded rate. No-arbitrage therefore requires $F_t \leq S_t e^{(r+u)\tau}$.

Crucially, the convenience yield does not enter the upper bound. The convenience yield is a benefit that accrues to the holder of physical inventory; it lowers the futures price relative to full carry and so can only relax the bound, never tighten it. A model-free arbitrage bound must not reference the latent state being inferred, and this bound does not. The corresponding loss, evaluated on the data manifold (32), is

$$\hat{\mathcal{L}}_{h,\text{cac}} = \frac{1}{N_t} \sum_{i=1}^{N_t} m_h^{(i)} \left[\max\left(0, \hat{F}_\theta(x_i) - S_i e^{(r+u)\tau_i}\right) \right]^2. \quad (34)$$

Taking logs of (33) and using the closed form $\hat{F} = S \exp[B(\tau)\delta + A(\tau)]$ with $B(\tau) < 0$ shows the bound is equivalently a lower bound on the convenience yield,

$$B(\tau)\delta + A(\tau) \leq (r+u)\tau \iff \delta \geq \frac{(r+u)\tau - A(\tau)}{B(\tau)}, \quad (35)$$

the inequality reversing because $B(\tau) < 0$. Under the free surface $\hat{F}_\theta$ this equivalence holds only approximately, so (35) is retained as an internal consistency check evaluated at validation time rather than as a training identity: the cash-and-carry price ceiling and the economic-plausibility floor of Section 7.3 both act to bound the inferred yield from below, so the two constraints reinforce rather than oppose one another.

7.2 Model-free constraint II: reverse cash-and-carry lower bound

The reverse strategy (short the physical commodity, invest the proceeds at r , and buy the futures) would appear to enforce a symmetric lower bound $F_t \geq S_t e^{r\tau}$. For commodities this symmetry fails, and the asymmetry is the defining economic feature of the market. To short the physical the arbitrageur must borrow the commodity from an inventory holder, and that holder forgoes the convenience yield for the duration of the loan; in a competitive market they demand compensation for it. The convenience yield is therefore a wedge that weakens, and in the limit removes, the lower bound. A futures price below $S_t e^{r\tau}$, a backwarddated curve, is not in general an arbitrage but the normal signature of a positive convenience yield.

The correct model-free statement carries the convenience-yield wedge explicitly. Writing $\delta_{\max}$ for the largest economically defensible convenience yield, the lower bound is

$$F_t \geq S_t e^{(r-\delta_{\max})\tau}, \quad (36)$$

with corresponding loss, again on the data manifold,

$$\hat{\mathcal{L}}_{h,\text{rcac}} = \frac{1}{N_t} \sum_{i=1}^{N_t} m_h^{(i)} \left[\max\left(0, S_i e^{(r-\delta_{\max})\tau_i} - \hat{F}_\theta(x_i)\right) \right]^2. \quad (37)$$

The naive bound $F_t \geq S_t e^{r\tau}$ is recovered only in the limit $\delta_{\max} \rightarrow 0$, i.e. for an asset with no convenience yield, where the commodity behaves like a financial forward. Because $\delta_{\max}$ is a calibrated plausibility ceiling rather than a quantity fixed by pure arbitrage, (37) sits at the boundary between the model-free and economic-plausibility families; we retain it here as the convenience-yield-corrected counterpart of the cash-and-carry bound.

7.3 Economic-plausibility constraint: the convenience-yield floor

The carry bounds of Sections 7.1–7.2 constrain the futures price. A separate constraint bounds the convenience yield directly from below. Unlike a volatility, the convenience yield may be negative: a commodity in deep contango, where storage cost dominates the benefit of holding inventory, carries $\delta_t < 0$. It cannot, however, fall without limit. Below some floor $\delta_{\min}$ a rational holder abandons the physical position rather than continue absorbing the net carrying cost, so the convenience yield observed in a market of willing holders is bounded below. The constraint acts on the latent path at observation times:

$$\hat{\delta}_\phi(t_i) \geq \delta_{\min}, \quad \hat{\mathcal{L}}_{h,\delta\text{-floor}} = \frac{1}{N_t} \sum_{i=1}^{N_t} m_h^{(i)} \left[\max\left(0, \delta_{\min} - \hat{\delta}_\phi(t_i)\right) \right]^2. \quad (38)$$

Both $\delta_{\min}$ and $\hat{\delta}_\phi$ are rates (dimension: time^{-1}), so the bound is dimensionally consistent and, being a property of the yield alone, carries no dependence on the price level S_t .

Setting $\delta_{\min}$. We fix $\delta_{\min}$ empirically rather than derive it. A decision-theoretic threshold, obtained by solving the holder’s optimal-stopping problem for the point of abandonment, was considered and rejected on grounds of internal consistency. Such a floor is a structural restriction, whereas δ_t here is a reduced-form Ornstein–Uhlenbeck factor whose Gaussian marginals place positive probability on every finite level. A hard structural floor would therefore contradict the very dynamics that the PDE residual $\hat{\mathcal{L}}_f$ enforces, and would bias $\hat{\delta}_\phi$ upward in exactly the deep-contango regime the model is intended to capture. We instead take $\delta_{\min}$ as the empirical lower edge of the convenience-yield range in the Monte Carlo simulated data,

$$\delta_{\min} = -0.3 \text{ yr}^{-1}, \quad (39)$$

so that the constraint regularises the inversion without binding on economically plausible states. Written out with this value, the active penalty is

$$\hat{\mathcal{L}}_{h,\delta\text{-floor}} = \frac{1}{N_t} \sum_{i=1}^{N_t} m_h^{(i)} \left[\max\left(0, -0.3 - \hat{\delta}_\phi(t_i)\right) \right]^2, \quad (40)$$

firing only where the inferred yield falls below -0.3 yr^{-1} and returning zero on all admissible states.

8 The composite objective

Training minimises a single scalar objective assembled from a data-misfit term, terms enforcing the model dynamics on the state space, and no-arbitrage inequality penalties:

$$\hat{\mathcal{L}} := \underbrace{\lambda_t \hat{\mathcal{L}}_t}_{\text{Data misfit}} + \underbrace{\lambda_b \hat{\mathcal{L}}_b + \lambda_f \hat{\mathcal{L}}_f}_{\text{PDE structure}} + \underbrace{\sum_{\alpha} \lambda_{h_\alpha} \hat{\mathcal{L}}_{h_\alpha}}_{\text{No-arbitrage penalties}}, \quad \alpha \in \{\text{cac}, \text{rcac}, \delta\text{-floor}\}. \quad (41)$$

The category weights $\lambda_t, \lambda_b, \lambda_f, \{\lambda_{h_\alpha}\}$ are set adaptively by the WamOL loss-balancing update, and each term carries pointwise self-adaptive weights $m^{(i)}$ updated by gradient ascent, so that no single term dominates the gradient purely by scale. In Hoshisashi et al. (2024) the data term is annotated as the loss on *increasing inputs*, reflecting the intraday streaming setting in which the observation set grows through the trading session under the time-decay weight ζ ; we retain that mechanism as an option in Section 9 and adopt the neutral label above.

9 Data-misfit loss $\hat{\mathcal{L}}_t$

Each observation i consists of a spot price S_i , an observation time t_i , a time to maturity τ_i , and an observed futures price F_i^{obs} . The model price is the network surface evaluated at the inferred state,

$$\hat{F}_i := \hat{F}_\theta(S_i, \hat{\delta}_\phi(t_i), \tau_i), \quad (42)$$

and the data-misfit loss is taken in log-price,

$$\hat{\mathcal{L}}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} m_t^{(i)} \zeta(t_i, t) \left(\ln \hat{F}_i - \ln F_i^{\text{obs}} \right)^2, \quad \zeta(t_i, t) = e^{-\mu(t-t_i)}, \quad (43)$$

where $m_t^{(i)}$ are the self-adaptive pointwise weights and ζ is the time-decay weight of the online-learning component; in a static historical calibration $\zeta \equiv 1$ and the term reduces to a weighted mean squared log-price error. Working in log-price preserves scale invariance across price regimes and matches the linear measurement equation of the Kalman-filter baseline. The residual is no longer linear in the latent yield, since $\hat{\delta}_\phi$ enters through the network surface rather than through the exponential-affine closed form; the gradient coupling $\partial \hat{\mathcal{L}}_t / \partial \phi$ flows through $\partial \hat{F}_\theta / \partial \delta$, so the data loss is the sole channel through which observations discipline the inferred path, and the PDE residual is the sole channel through which the dynamics discipline the surface.

10 PDE-residual loss $\hat{\mathcal{L}}_f$

The futures pricing PDE (19) is recast as a residual operator acting on the network surface, with every coefficient evaluated at the mesh coordinates (S, δ, τ) :

$$r_{\text{PDE}}[\hat{F}_\theta](S, \delta, \tau) := \frac{\partial \hat{F}_\theta}{\partial \tau} - (r - \delta)S \frac{\partial \hat{F}_\theta}{\partial S} - \kappa(\alpha^\mathbb{Q} - \delta) \frac{\partial \hat{F}_\theta}{\partial \delta} - \frac{1}{2} \sigma_1^2 S^2 \frac{\partial^2 \hat{F}_\theta}{\partial S^2} - \rho \sigma_1 \sigma_2 S \frac{\partial^2 \hat{F}_\theta}{\partial S \partial \delta} - \frac{1}{2} \sigma_2^2 \frac{\partial^2 \hat{F}_\theta}{\partial \delta^2}. \quad (44)$$

The latent path $\hat{\delta}_\phi$ does not appear: the PDE is a statement about the pricing function on the whole state space, and its drift coefficients are functions of the coordinate δ at which the derivatives are taken. All partial derivatives are obtained by automatic differentiation of the network. Evaluating on N_f collocation points sampled over (S, δ, τ) ,

$$\hat{\mathcal{L}}_f = \frac{1}{N_f} \sum_{j=1}^{N_f} m_f^{(j)} \left(r_{\text{PDE}}[\hat{F}_\theta](x^{(j)}) \right)^2. \quad (45)$$

A surface consistent with the Gibson–Schwartz dynamics drives this residual to zero at every collocation point; deviations are penalised quadratically.

11 Initial-condition loss $\hat{\mathcal{L}}_b$

The change of variables to time-to-maturity converts the terminal condition into the initial condition (20), enforced on N_b points sampled over the $\tau = 0$ slice of the mesh:

$$\hat{\mathcal{L}}_b = \frac{1}{N_b} \sum_{j=1}^{N_b} m_b^{(j)} \left(\hat{F}_\theta(S_j, \delta_j, 0) - S_j \right)^2. \quad (46)$$

Remark (Hard enforcement). *The initial condition and positivity can instead be imposed exactly through the output transform*

$$\hat{F}_\theta(S, \delta, \tau) = S \exp(\tau N_\theta(S, \delta, \tau)), \quad (47)$$

where N_θ is the raw network output. At $\tau = 0$ the condition holds identically, $\hat{\mathcal{L}}_b$ vanishes by construction and is dropped from the objective (41), mirroring the omission of $\mathcal{L}_b$ in Hoshisashi et al., where the softplus output activation renders the boundary conditions automatic. The transform also makes the log-price used in $\hat{\mathcal{L}}_t$ linear in the raw output, $\ln \hat{F}_\theta = \ln S + \tau N_\theta(S, \delta, \tau)$, which conditions the inversion well and recovers, in a different guise, the linearity that the exponential-affine closed form provided in the earlier formulation.

12 WamOL weight-update mechanics

Collect the trainable parameters as $\psi := (\theta, \phi)$ and index the loss categories by $\beta \in \mathcal{B} := \{t, b, f, \text{cac}, \text{rcac}, \delta\text{-floor}\}$. Every category loss in Sections 7 to 11 has the common form

$$\hat{\mathcal{L}}_\beta(m_\beta, \psi) = \frac{1}{N_\beta} \sum_{i=1}^{N_\beta} m_\beta^{(i)} \zeta_i (e_\beta^{(i)}(\psi))^2, \quad m_\beta^{(i)}(0) = 1, \quad (48)$$

where $\zeta_i = \zeta(t_i, t)$ for the time-indexed terms and $\zeta_i \equiv 1$ otherwise, and the pointwise residuals are

$$\begin{aligned} e_t^{(i)} &= \ln \hat{F}_\theta(x_i) - \ln F_i^{\text{obs}}, & e_b^{(j)} &= \hat{F}_\theta(S_j, \delta_j, 0) - S_j, & e_f^{(j)} &= r_{\text{PDE}}[\hat{F}_\theta](x^{(j)}), \\ e_{\text{cac}}^{(i)} &= \max(0, \hat{F}_\theta(x_i) - S_i e^{(r+u)\tau_i}), & e_{\text{rcac}}^{(i)} &= \max(0, S_i e^{(r-\delta_{\text{max}})\tau_i} - \hat{F}_\theta(x_i)), & e_{\delta\text{-fl}}^{(i)} &= \max(0, \delta_{\text{min}} - \hat{\delta}_\phi(t_i)), \end{aligned} \quad (49)$$

with $x_i = (S_i, \hat{\delta}_\phi(t_i), \tau_i)$ on the data manifold and $x^{(j)}$ on the mesh. Each squared hinge is continuously differentiable despite the max, since $\frac{d}{dz} \max(0, z)^2 = 2 \max(0, z)$ is continuous through the kink; no subgradient treatment is needed. The composite objective is $\hat{\mathcal{L}}(\lambda, m, \psi) = \sum_\beta \lambda_\beta \hat{\mathcal{L}}_\beta(m_\beta, \psi)$, and training is the saddle problem

$$\min_\psi \max_m \hat{\mathcal{L}}(\lambda, m, \psi), \quad (50)$$

with λ set by the balancing rule of Section 12.2. The two weight families factorise cleanly: m allocates emphasis *within* a category, λ allocates it *across* categories.

12.1 First whack: self-adaptive pointwise weights

Because (48) is linear in each $m_\beta^{(i)}$, the ascent gradient is available in closed form,

$$\nabla_{m_\beta^{(i)}} \hat{\mathcal{L}}_\beta = \frac{\zeta_i}{N_\beta} (e_\beta^{(i)})^2 \geq 0, \quad (51)$$

so the update at every q_m -th epoch is

$$m_\beta^{(i)}(p+1) = m_\beta^{(i)}(p) + \frac{\eta_m \zeta_i}{N_\beta} (e_\beta^{(i)}(p))^2. \quad (52)$$

Three consequences follow. First, the increments are nonnegative, so each weight is monotone nondecreasing from its initialisation $m = 1$ and admits the cumulative reading

$$m_\beta^{(i)}(p) = 1 + \frac{\eta_m \zeta_i}{N_\beta} \sum_{\substack{q < p \\ q \equiv 0 \pmod{q_m}}} (e_\beta^{(i)}(q))^2 : \quad (53)$$

a point's weight is its integrated squared-residual history, and emphasis accumulates precisely where misfit persists across epochs. Second, for the hinge terms the increment vanishes on every satisfied point, so weight inflates only while a point is in violation and freezes the moment it is whacked; this is the mechanism by which sparse, intermittent constraint violations are not drowned out by the dense smooth losses. Third, the constraint residuals of (49) are evaluated at moving states: x_i shifts during training through $\hat{\delta}_\phi(t_i)$, so the weight $m_h^{(i)}$ attaches to the *observation index* rather than to a fixed location in state space, in contrast to Hoshisashi et al., where the constraint points x_h are fixed mesh grids.

Remark (Maturity-dependent identifiability). *The coupling of the data residual to the latent path is*

$$\frac{\partial e_t^{(i)}}{\partial \hat{\delta}_\phi(t_i)} = \left. \frac{\partial \ln \hat{F}_\theta}{\partial \delta} \right|_{x_i}, \quad (54)$$

which near the true solution behaves as $B(\tau_i) = -(1 - e^{-\kappa\tau_i})/\kappa$: it vanishes linearly as $\tau_i \rightarrow 0$ and saturates at $-1/\kappa$ for long maturities. Short-dated contracts therefore carry almost no gradient information about ϕ , and the yield is structurally identified by the term-structure slope across maturities observed at the same t_i . The self-adaptive weights partially compensate for the resulting systematic variation in residual scale across the curve, but they cannot manufacture identification where $B(\tau_i) \approx 0$; a calibration window should therefore always contain contracts of sufficient tenor.

12.2 Second whack: category loss balancing

The categories touch different parameter subsets:

$$\psi_t = \psi_{\text{cac}} = \psi_{\text{rcac}} = (\theta, \phi), \quad \psi_b = \psi_f = \theta, \quad \psi_{\delta\text{-floor}} = \phi. \quad (55)$$

Define each category's gradient scale on its own support,

$$\bar{g}_\beta(p) := \overline{|\nabla_{\psi_\beta} \hat{\mathcal{L}}_\beta(p)|}, \quad (56)$$

the mean of the absolute values of the entries. The restriction to ψ_β matters in the two-network setting: taking the mean over the full vector ψ would dilute $\bar{g}_{\delta\text{-floor}}$ by the factor $\dim \phi / \dim \psi$, since every θ -entry of its gradient is identically zero, and the balancing rule below would inflate $\lambda_{\delta\text{-floor}}$ by the reciprocal factor. With $\dim \phi \ll \dim \theta$ (a scalar path network against a surface network) the distortion would be one to two orders of magnitude; evaluating (56) on the support removes it. The balancing update at every q_λ -th epoch is

$$\lambda_\beta(p+1) = \begin{cases} 1, & \bar{g}_\beta(p) = 0, \\ \frac{1}{2} \left(\lambda_\beta(p) + \frac{G(p)}{\bar{g}_\beta(p)} \right), & \text{otherwise,} \end{cases} \quad G(p) := \sum_{\beta' \in \mathcal{B}} \bar{g}_{\beta'}(p). \quad (57)$$

Fixed point and convergence. Freeze the gradient scales, $\bar{g}_\beta(p) \equiv \bar{g}_\beta$, and write $T_\beta := G/\bar{g}_\beta$. The recursion (57) is affine, $\lambda_\beta(p+1) = \frac{1}{2}\lambda_\beta(p) + \frac{1}{2}T_\beta$, with solution

$$\lambda_\beta(p) = T_\beta + 2^{-p}(\lambda_\beta(0) - T_\beta), \quad (58)$$

so λ_β converges geometrically to T_β at ratio $\frac{1}{2}$ per balancing event. At the fixed point,

$$\lambda_\beta \bar{g}_\beta = G \quad \text{for every } \beta \in \mathcal{B} : \quad (59)$$

each category contributes the same mean absolute gradient magnitude to $\nabla_\psi \hat{\mathcal{L}}$, so no term dominates the descent direction purely by scale. This is the property the mechanism buys; the factor $\frac{1}{2}$ is a damping choice trading tracking speed against oscillation when the $\bar{g}_\beta$ drift between updates.

Hinge reset. Once a constraint holds on all of its evaluation points, its squared hinge is identically zero on a neighbourhood of the current parameters, so $\bar{g}_\beta = 0$ *exactly* and (57) resets $\lambda_\beta \rightarrow 1$. The reset is harmless: a locally flat loss contributes no gradient at any finite weight. In market data free of realised arbitrage this is the expected late-training state of both carry penalties, so persistent balancing operates mainly among $\{t, f, b\}$; under hard enforcement of the initial condition (Remark, Section 11) the set shrinks to $\{t, f\}$ and b is removed from $\mathcal{B}$, mirroring the omission of $\mathcal{L}_b$ in the reference implementation. The use of absolute rather than squared values in (56) follows the same logic: constraint gradients are sparse late in training, and a squared mean would suppress the few active entries that remain.

12.3 Third whack: online incremental learning

The streaming component rests on a single property of the decay kernel.

Remark (Semigroup property). *For $t_i \leq t_1 \leq t_2$ the exponential kernel factorises,*

$$\zeta(t_i, t_2) = e^{-\mu(t_2-t_i)} = e^{-\mu(t_2-t_1)} e^{-\mu(t_1-t_i)} = \zeta(t_1, t_2) \zeta(t_i, t_1), \quad (60)$$

and among kernels of the form $\zeta(t_i, t) = \varphi(t - t_i)$ it is the only measurable solution of $\varphi(a+b) = \varphi(a)\varphi(b)$.

Split the observation set at t_1 into old points ($t_i \leq t_1$, N_{t_1} of them) and new points ($t_1 < t_i \leq t_2$). Applying (60) to the old block of (48):

$$\begin{aligned} \hat{\mathcal{L}}_{t_2} &= \frac{1}{N_{t_2}} \left[\zeta(t_1, t_2) \sum_{i \leq N_{t_1}} m_t^{(i)} \zeta(t_i, t_1) (e_t^{(i)})^2 + \sum_{i > N_{t_1}} m_t^{(i)} \zeta(t_i, t_2) (e_t^{(i)})^2 \right] \\ &= \frac{N_{t_1}}{N_{t_2}} \zeta(t_1, t_2) \hat{\mathcal{L}}_{t_1} + \frac{N_{t_2} - N_{t_1}}{N_{t_2}} \hat{\mathcal{L}}_{\Delta t}, \end{aligned} \quad (61)$$

recovering the recursive form of Hoshisashi et al. eq. (25). The derivation shows why the exponential is not a modelling convenience but a computational requirement: the entire history enters through a single stored aggregate rescaled by $\zeta(t_1, t_2)$, whereas any non-exponential decay breaks the factorisation and forces every historical residual to be reweighted individually at each arrival. The identity is exact for fixed residuals; in training the residuals move with ψ , so (61) is used as the definition of the streamed objective evaluated at the current parameters.

Scope in this problem. Calendar time enters the formulation only through the observation set and the latent path: the surface $\hat{F}_\theta$ is time-homogeneous because the PDE coefficients are constant. The streaming decomposition therefore applies to $\hat{\mathcal{L}}_t$ and, if the same ζ is adopted, to the data-manifold penalties $\hat{\mathcal{L}}_{h,\text{cac}}$, $\hat{\mathcal{L}}_{h,\text{rcac}}$, and $\hat{\mathcal{L}}_{h,\delta\text{-floor}}$; the mesh losses $\hat{\mathcal{L}}_f$ and $\hat{\mathcal{L}}_b$ carry no time index and are excluded. A further asymmetry with the implied-volatility case concerns the trigger. A new observation date extends the required domain of $\hat{\delta}_\phi$ beyond its trained support, where neural extrapolation is unreliable, so the recalibration event (fired when $\hat{\mathcal{L}}_{\Delta t} > \varepsilon$) doubles as the mechanism by which the path is extended and can be expected to fire on most arrival dates in a daily-settlement setting; in the intraday IVS case the surface interpolates in t and arrivals frequently pass without recalibration. With daily data, μ has units of inverse trading days and sets the memory half-life $\ln 2/\mu$; a static historical calibration is recovered as $\mu \rightarrow 0$, giving $\zeta \equiv 1$.

12.4 Assembled update loop

Initialise $m_\beta^{(i)} = 1$, $\lambda_\beta = 1$, and $\psi = (\theta, \phi)$; fix η , η_m , q_m , q_λ , $p_{\max}$, μ , ε . Then, for $p = 1, \dots, p_{\max}$:

1. Evaluate every $\hat{\mathcal{L}}_\beta$ from (48) and compute $\nabla_{\psi_\beta} \hat{\mathcal{L}}_\beta$ by automatic differentiation on the supports (55).
2. If $p \equiv 0 \pmod{q_m}$: ascend the pointwise weights by (52).
3. If $p \equiv 0 \pmod{q_\lambda}$: rebalance the category weights by (57)–(56).
4. Descend the parameters jointly, $\psi(p+1) = \psi(p) - \eta \nabla_\psi \sum_\beta \lambda_\beta \hat{\mathcal{L}}_\beta(p)$.

On arrival of new observations at t_2 : form $\hat{\mathcal{L}}_{\Delta t}$ on the new points; if $\hat{\mathcal{L}}_{\Delta t} > \varepsilon$, assemble the streamed data loss by (61), extend the evaluation sets of the data-manifold penalties to the new indices, reset $p \leftarrow 1$, and resume the loop; otherwise continue without recalibration.